

Computer Networks

Lecture 33 Application Layer

Content



- **DNS**
- **HTTP**
- **SMTP**

Introduction

what happens when a user opens a website?

URL entered

DNS lookup

TCP/TLS setup

HTTP request

Response rendered

- The browser first needs the server address
- Then it creates a connection
- Sends an HTTP message
- Receives resources and renders the page

**DNS answers “where is it?” while
HTTP/HTTPS answers “how do we exchange
web data?”**

URL: locating a web resource

`https://www.example.com:443/path/page.html?course=cn#section1`

Scheme
`https`

Host name
`www.example.com`

Port
`443`

Path
`/path/page.html`

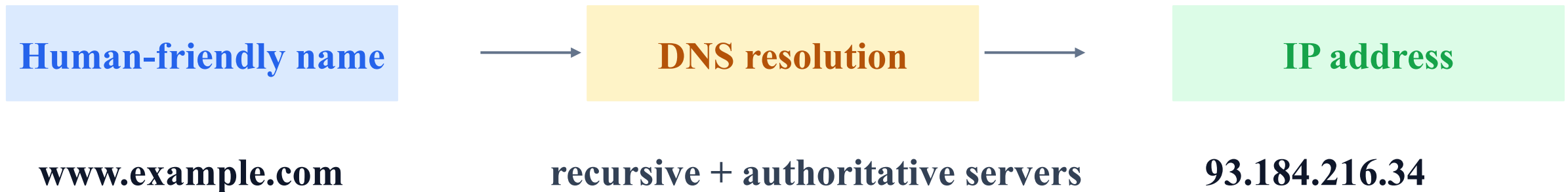
Query
`?course=cn`

Fragment
`#section1`

The host name is resolved by DNS. The scheme decides whether the browser uses plain HTTP or HTTPS over TLS.

DNS

- DNS maps domain names to IP addresses and other service information.
- It is distributed and hierarchical, not a single central table.
- The result is cached to reduce delay and DNS traffic.



DNS: Name Space

Name Space means the system used to give names to machines and map those names to their IP addresses. Since IP addresses are unique, machine names should also be unique to avoid confusion

There are two types of name spaces: Flat Name Space and Hierarchical Name Space

Flat Name Space: names are simple strings without any structure.

Example: challenger

The problem is that it needs central control to make sure no two machines get the same name. So, it is not suitable for a large system like the Internet.

DNS: Name Space

Hierarchical Name Space: In a hierarchical name space, names are divided into parts.

Example: challenger.berkeley.edu

Here:

- *edu* shows the type of organization
- *berkeley* shows the organization name
- *challenger* is the machine/host name

This system is better because name management can be **decentralized**. A central authority controls only the main part, such as *.edu* or *.com*, while each organization controls its own internal names.

DNS: Name Space

Fully Qualified Domain Name: An FQDN gives the complete name of a host from the specific host up to the root.

Example: challenger.atc.fhda.edu.

It ends with a dot, which represents the root.

Partially Qualified Domain Name: A PQDN is an incomplete domain name.

Example: challenger

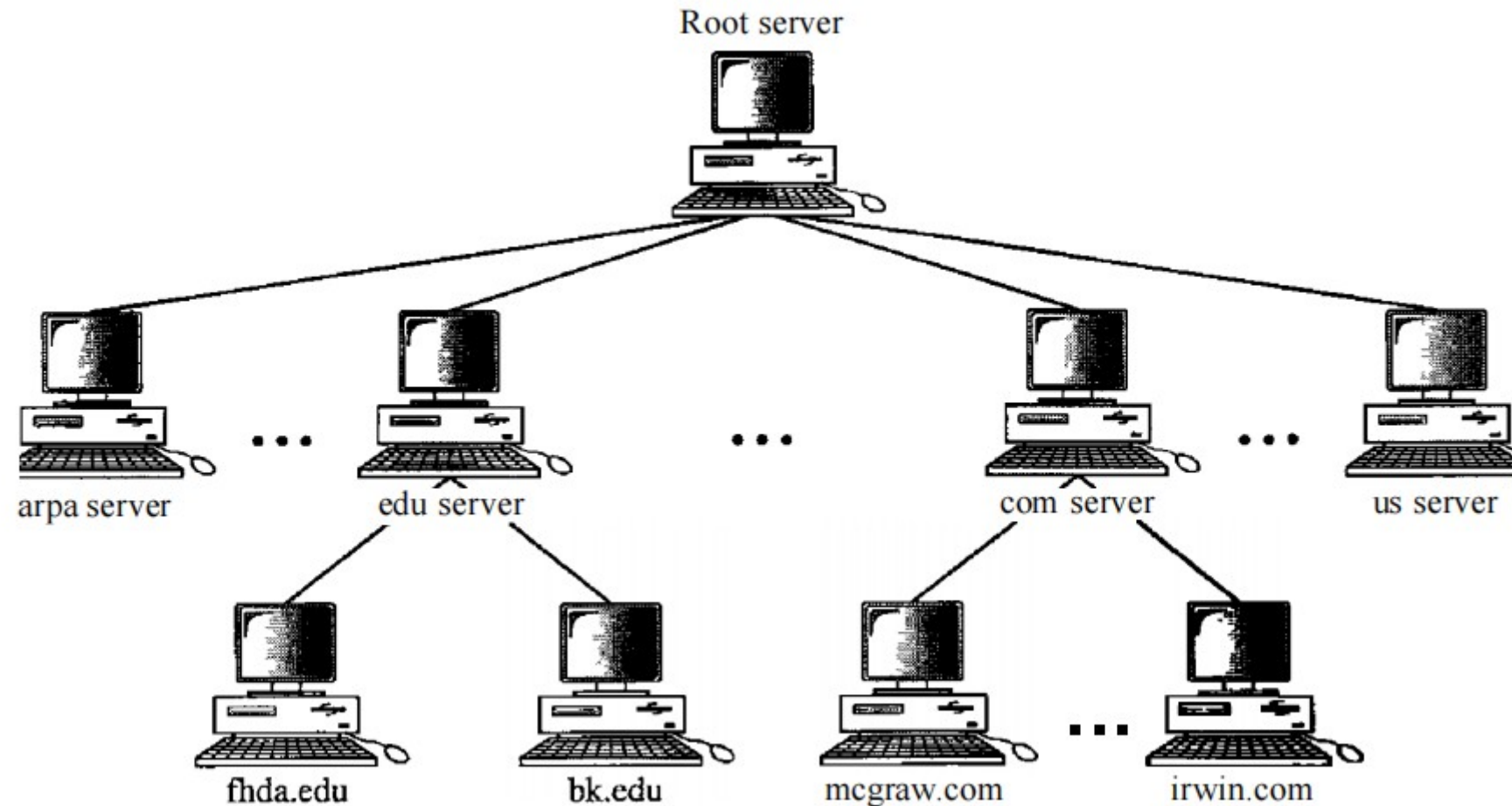
The DNS client adds the missing suffix, such as: atc.fhda.edu.

So, challenger becomes: challenger.atc.fhda.edu.

DNS: Name Space

Distribution of Name Space

It is not practical to store the entire DNS database on one computer. So, DNS information is distributed among many computers called DNS servers.



DNS: Name Space

Zone

Since the complete domain name hierarchy cannot be stored on a single server, it is divided among many servers.

What a server is responsible for or has authority over is called a zone.

If a server accepts responsibility for a domain and does not divide the domain into smaller domains, the *domain* and the *zone* refer to the same thing. The server makes a database called a *zone file* and keeps all the information for every node under that domain.

However, if a server divides its domain into subdomains and delegates part of its authority to other servers, *domain* and *zone* refer to different things.

DNS: Name Space

Root Server

A root server is a server whose zone consists of the whole tree. A root server usually does not store any information about domains but delegates its authority to other servers, keeping references to those servers

Primary Server

A primary DNS server stores and maintains the original zone file. It can create, update, and modify DNS records.

Secondary Server

A secondary DNS server gets a copy of the zone file from the primary server. This process is called zone transfer. Both primary and secondary servers are authoritative, but the secondary server mainly provides backup and redundancy.

DNS: Resolution

Mapping a name to an address or an address to a name is called name-address resolution

A host uses a DNS client called a **resolver** to find:

- name to address mapping, or
- address to name mapping.

- The resolver sends the request to the nearest DNS server.
- If the DNS server has the required information, it sends the answer back.
- If the server does not have the information, it:
 - refers the resolver to another DNS server, or
 - contacts other servers to get the information.
- After receiving the response, the resolver checks whether it is:
 - a valid answer, or
 - an error.
- Finally, the resolver sends the result to the application/process that requested it.

DNS: Resolution

Mapping Names to Addresses

Most of the time, the resolver gives a domain name to the server and asks for the corresponding address. In this case, the server checks the domains to find the mapping.

Mapping Addresses to Names

It is used to find the domain name from an IP address.

Domain name → IP address

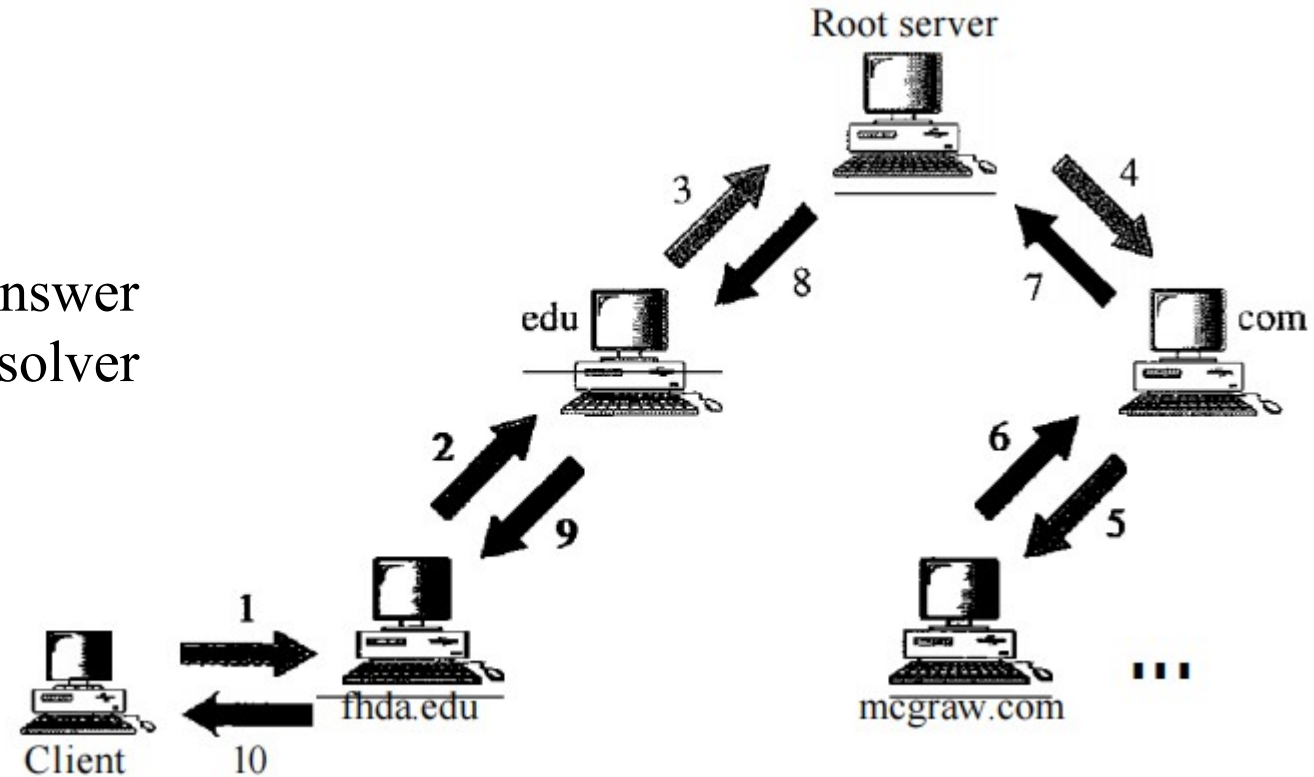
Mail servers use reverse DNS to check whether the sender's IP address has a valid domain name. This helps reduce spam.

- If a server sees a request from: 142.250.182.14
- Reverse DNS can help identify it as belonging to a Google server/domain.

DNS: Resolution

Recursive Resolution

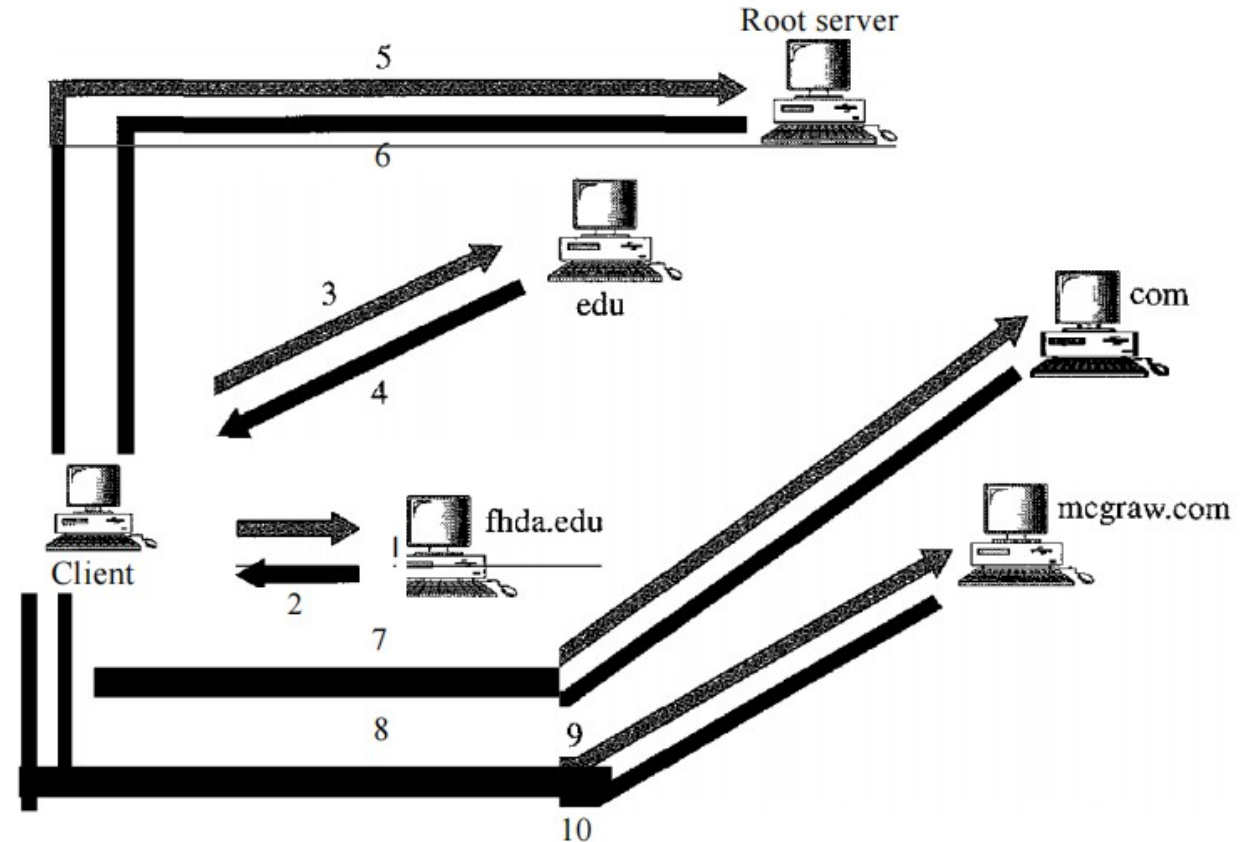
The client (resolver) can ask for a recursive answer from a name server. This means that the resolver expects the server to supply the final answer.



DNS: Resolution

Iterative Resolution

If the client does not ask for a recursive answer, the mapping can be done iteratively. If the server is an authority for the name, it sends the answer. If it is not, it returns (to the client) the IP address of the server that it thinks can resolve the query. The client is responsible for repeating the query to this second server.



Electronic Mail

One of the most popular Internet services is electronic mail (e-mail).

To explain the architecture of e-mail, we explain scenarios from simplest arrangement to the most common email exchange.

1. Same mail server

Sender and receiver share one mail server
Only two User Agents (UAs) are needed

2. Different mail servers

Sender and receiver use different servers
SMTP works between client and server
MTAs

3. Sender remote

Sender connects to own mail server
through LAN/WAN. Two pairs of MTAs
are involved

4. Both remote

Both users access their mail servers
remotely Uses UA + SMTP/MTA +
POP3/IMAP MAA

Scenario 1: Sender and Receiver on Same Mail Server



Key idea

- Both users are registered on the same mail server.
- No inter-server mail transfer is required.
- Only the two user agents are needed to compose and read email.

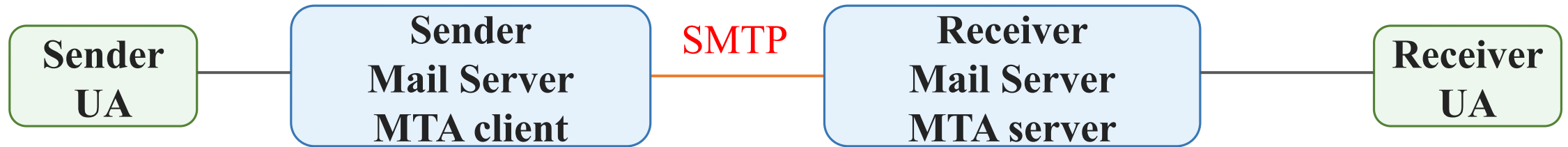
User Agent

- Email application used by a person
- Examples: Gmail app, Outlook, Thunderbird
- Used to compose, read and organize emails

Mail Server

- Stores user mailboxes
- Sends and receives messages
- Communicates with other mail servers

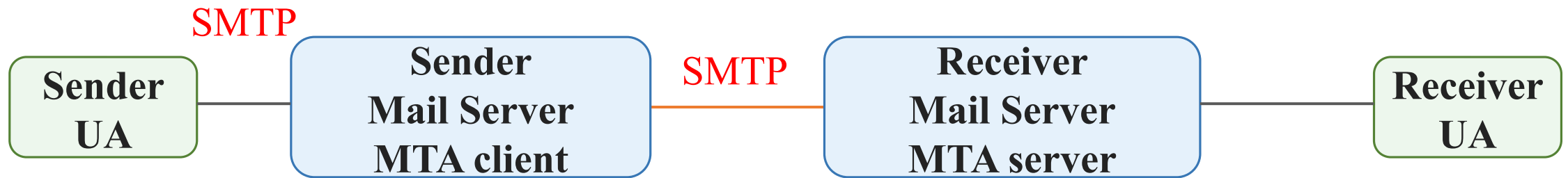
Scenario 2: Sender and Receiver on Different Mail Server



Key idea

- The sender and receiver belong to different mail servers.
- Mail transfer between the two servers is handled by SMTP.
- Two UAs and one client–server pair of MTAs are required.

Scenario 3: Sender Connected to Mail Server Through LAN/WAN



Key idea

- The sender is not directly on the mail server; sender reaches it through LAN/WAN.
- Mail needs SMTP first from sender side to its server, and then between servers.
- This requires two UAs and two client–server pairs of MTAs.

Scenario 4: Both Sender and Receiver Access Mail Remotely



Key idea

- Both users access their mail servers through a network connection.
- SMTP pushes the message between MTAs.
- POP3/IMAP pulls/accesses mail from the receiver's mailbox.
- This is the most common situation today.

SMTP Protocol

SMTP is used to send email messages from client to sender mail server or transfers mail between mail servers. It is called a push protocol because the sender pushes the message forward. Common ports: 25 for server-to-server, 587 for authenticated submission.

Characteristics of SMTP Protocol

- SMTP only sends or transfers email; it does not manage inbox viewing
- SMTP works reliably with store-and-forward communication
- If receiver server is not available, sender server may retry later
- Attachments are encoded before transmission
- Modern SMTP usually uses authentication and encryption for security

SMTP Working

SMTP is used to send email messages from client to sender mail server or transfers mail between mail servers. It is called a push protocol because the sender pushes the message forward. Common ports: 25 for server-to-server, 587 for authenticated submission

The basic conversation between client and server involves following command

- HELO/EHLO: client introduces itself to the SMTP server
- MAIL FROM: identifies the sender address
- RCPT TO: identifies the receiver address
- DATA: sends the message content
- QUIT: closes the SMTP session

POP3: Post Office Protocol Version 3

POP3 is used by a user agent to retrieve emails from a mail server. It usually downloads messages to the user's device



The basic Characteristics of POP3 are

- After download, messages may be deleted from the server depending on settings
- Simple protocol with limited mailbox management
- Suitable when email is accessed mainly from one device

POP3 Working

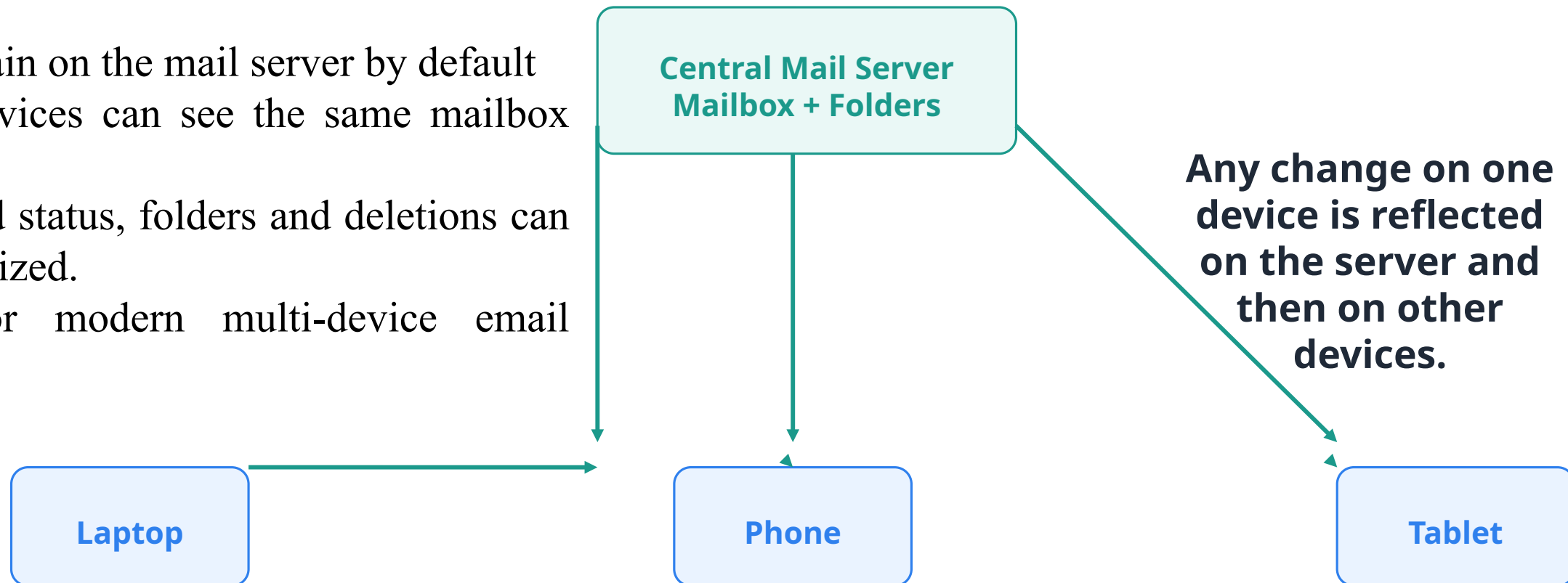
POP3 is used by a user agent to retrieve emails from a mail server. It usually downloads messages to the user's device



- Authorization: user logs in with username and password
- Transaction: client lists, retrieves or deletes messages
- Update: server applies delete operations and closes session

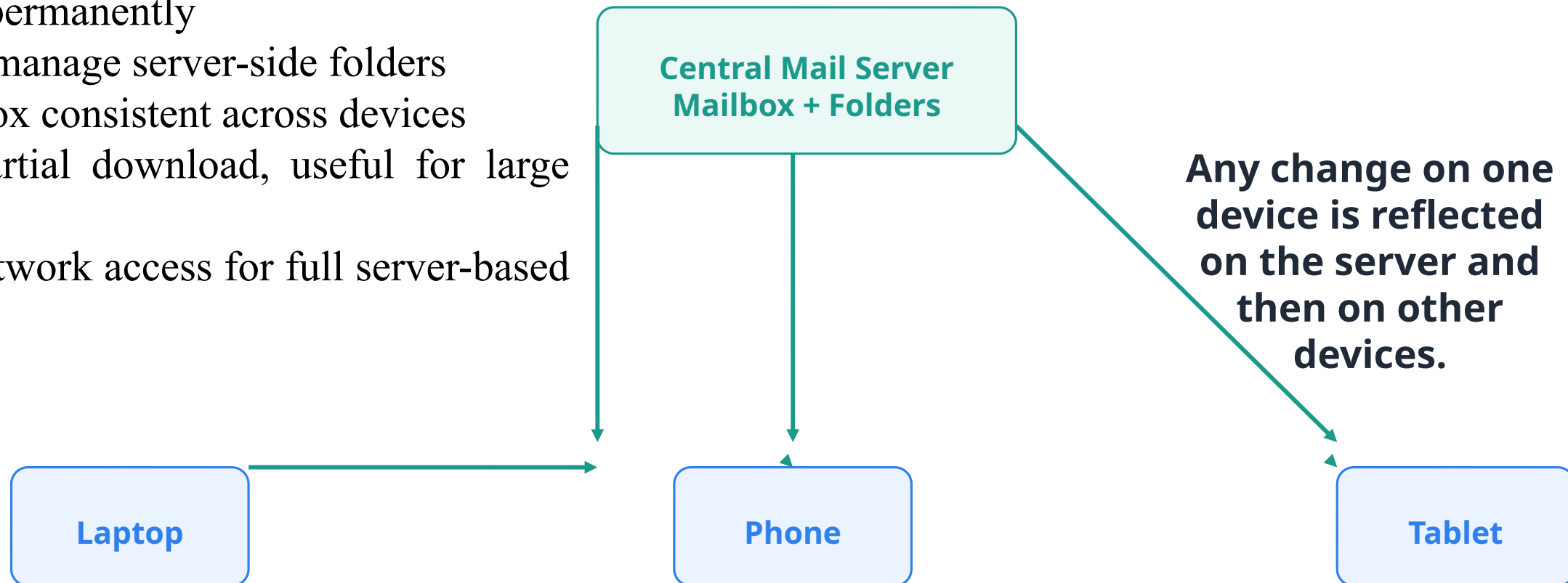
IMAP: Internet Message Access Protocol

- IMAP allows users to access email stored on the server.
- Emails remain on the mail server by default
- Multiple devices can see the same mailbox state.
- Read/unread status, folders and deletions can be synchronized.
- Suitable for modern multi-device email usage.



IMAP: Characteristics

- View messages without downloading everything permanently
- Search and manage server-side folders
- Keep mailbox consistent across devices
- Supports partial download, useful for large attachments
- Requires network access for full server-based operations



POP3 vs IMAP

Both are used to receive/access email, but they behave differently.

POP3

- Downloads email to device
- Simpler and older style
- Best for single-device use
- Limited synchronization
- Can save server storage if messages are removed

IMAP

- Keeps email on server
- Modern and flexible
- Best for multiple devices
- Synchronizes status and folders
- Needs more server storage

POP3 vs IMAP vs SMTP

Protocol	Main Use	Direction	Best Description
SMTP	Sending email	Client → mail Server → server	Outgoing mail protocol
POP3	Receiving email	Mail server → user device	Download-and-read model
IMAP	Accessing email	User device ↔ mail server	Server-sync model

MIME in mail Service

MIME allows email to support attachments, multimedia content, formatted text, and non-ASCII characters. Without MIME, email would be limited mainly to plain text only.

Traditional email could send only plain text messages. But modern users need to send files such as images, documents, and multimedia. So, MIME tells the mail system:

1. What type of data is inside the email.
2. How the data is encoded.
3. How the receiver should display or process it.

Example

When you send an email with a PDF attachment, MIME adds information such as:

Content-Type: application/pdf

Content-Transfer-Encoding: base64

Content-Disposition: attachment

This tells the receiver's mail client:

- The attachment is a PDF file
- It is encoded safely for transmission
- It should be shown as an attachment

Security

- Authentication verifies the user before sending or accessing mail
- TLS encryption protects communication between client and server
- Spam filtering checks suspicious email sources and content
- Reverse DNS and server reputation may be used in mail verification
- Users should avoid opening unknown attachments or links